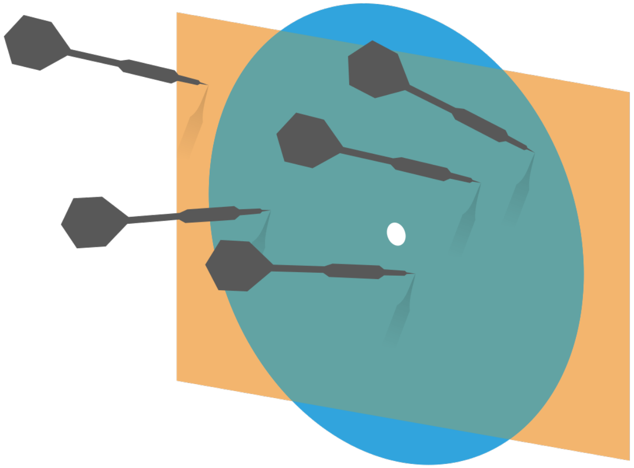
Probability and Statistics for Computer Science



Credit: wikipedia

Who discovered this?

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

Hongye Liu, Teaching Assistant Prof, CS361, UIUC, 09.26.2024

Last time

- Random Variable
  - Variance, *Covariance*
  - *The weak law of large numbers*

$$E[(X-E[X])^2] = E[X^2] - (E[X])^2$$

$$cov(X,Y) = E[(X-E[X])(Y-E[Y])]$$

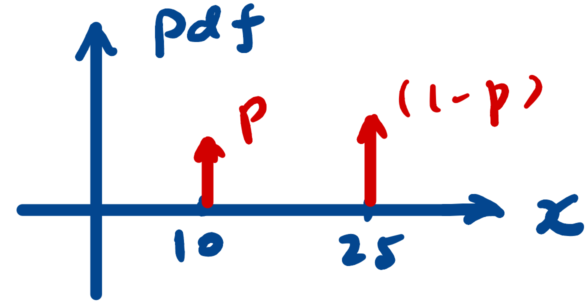
$$= E[XY] - E[X]E[Y] = \sigma_x \cdot \sigma_y \cdot corr(X,Y)$$

Random Variable Example for WLLN

Money box

10¢ 25¢

dime quarter



* shake and take one

and put back

(with replacement)

independent

$X_1 = 10?\ 25?$

$X_2 = 10?\ 25?$

$\vdots$

X_n

$E[X] = ?$

$10p + 25(1-p)$

$$\overline{X_n} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Proof of Weak law of large numbers

- Apply Chebyshev's inequality

$\forall \epsilon > 0$

$$P(|\overline{\mathbf{X}} - E[\overline{\mathbf{X}}]| \geq \epsilon) \leq \frac{var[\overline{\mathbf{X}}]}{\epsilon^2}$$

- Substitute $E[\overline{\mathbf{X}}] = E[X]$ and $var[\overline{\mathbf{X}}] = \frac{var[X]}{N}$

$$P(|\overline{\mathbf{X}} - E[\mathbf{X}]| \geq \epsilon) \leq \frac{var[\mathbf{X}]}{N\epsilon^2} \xrightarrow[N \to \infty]{} 0$$

$$\boxed{\lim_{N\to\infty} P(|\overline{\mathbf{X}} - E[X]| \geq \epsilon) = 0 \quad \lim_{N\to\infty} P(|\overline{X} - E[X]| < \epsilon) = 1}$$

$\forall \epsilon > 0$

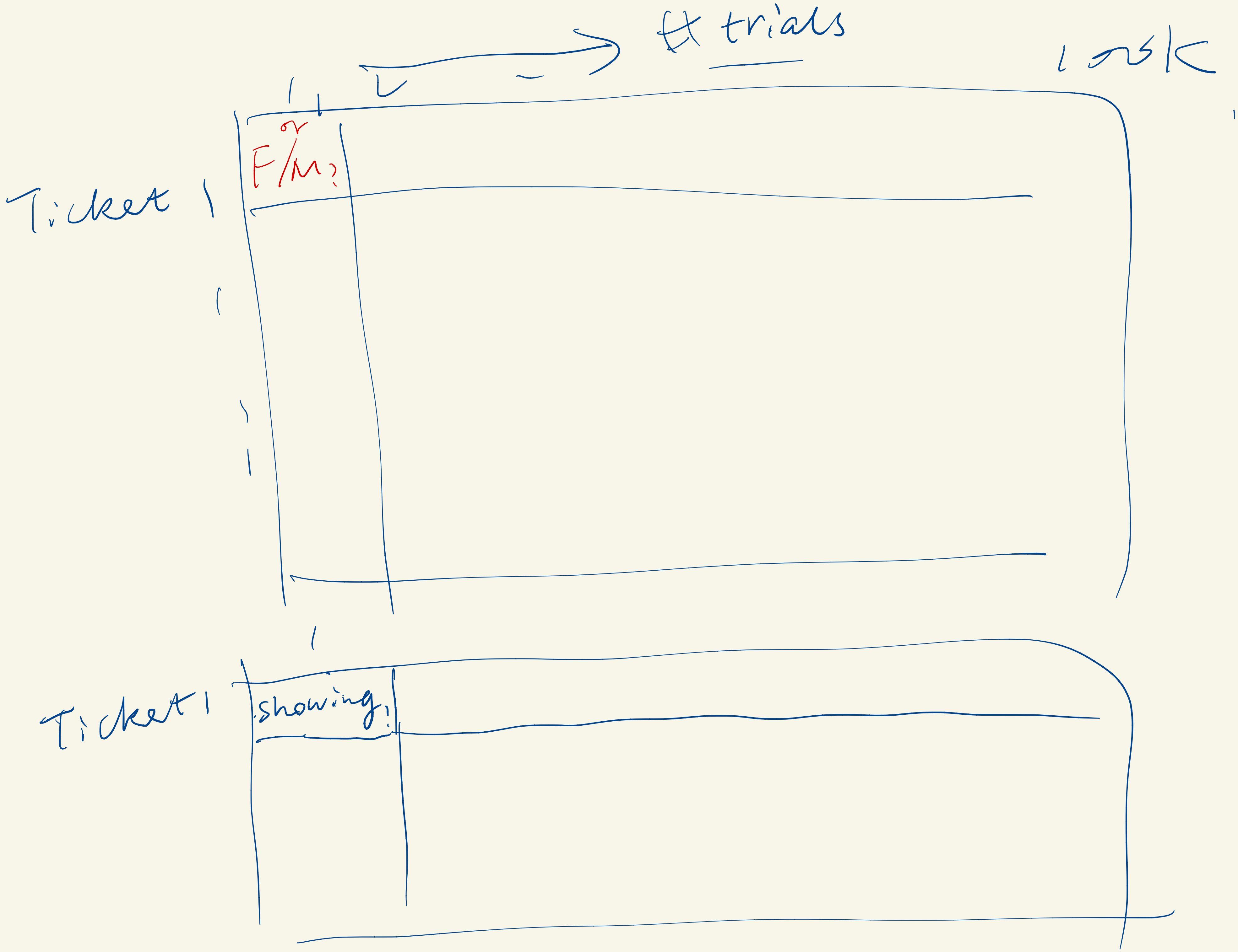
Applications of the Weak law of large numbers

- The law of large numbers ***justifies using simulations*** (instead of calculation) to estimate the expected values of random variables

$$\lim_{N\to\infty} P(|\overline{\mathbf{X}} - E[X]| \geq \epsilon) = 0$$

- The law of large numbers also ***justifies using histogram*** of large random samples to approximate the probability distribution function , see proof on

  Pg. 353 of the textbook by DeGroot, et al.



trials
task
Ticket 1
F or M?
Ticket 1
showing?

Objectives

- Important known discrete probability distributions
- Continuous Random Variable

The classic discrete distributions

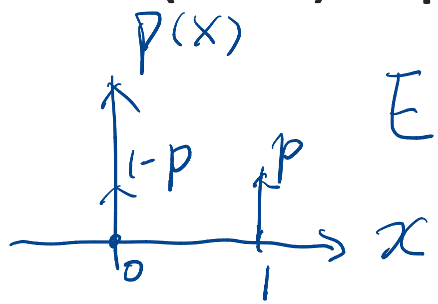
- Bernoulli X_{bei}
- Binomial $X_b = \sum_{i=1}^{N} X_{bei}$ X_{bei} are mutually indpt.
- Geometric
- Discrete uniform

Bernoulli distribution

- A random variable X is **Bernoulli** if it takes on two values 0 and 1 such that P(X=1) = p, P(X=0)=1-p

$$X(\omega) = \begin{cases} 1 & E_A \\ 0 & E_A^c \end{cases}$$

Indicator func.



$$E[X] = \sum_x x \cdot p(x)$$

$$= 1 \cdot p + 0 \times (1-p)$$

$$E[X] = p$$

$$var[X] = p(1-p)$$

$$\hookrightarrow E[X^2] - (E[X])^2$$

Jacob Bernoulli (1654-1705)

Credit: wikipedia

$$
\begin{aligned}
\operatorname{Var}[X] &= E[X^2] - (\underline{E[X]})^2 \\
&= \sum_x x^2 p(x) - p^2 \\
&= 0^2 \cdot (1-p) + 1^2 \cdot p - p^2 \\
&= p - p^2 \\
&= p(1-p)
\end{aligned}
$$

Bernoulli distribution

- Examples
  - Tossing a biased (or fair) coin
  - Making a free throw
  - Rolling a six-sided die and checking if it shows 6
  - **Any indicator function** of a random variable

$$X = \begin{cases} 1 & \text{Event A } (E_A) \text{ happens} \\ 0 & \text{ow} \end{cases}$$

$$E[X] = ? \quad 1 \times P(E_A) + 0\,(1 - P(E_A)]$$
$$= P(E_A)$$

Binomial distribution

- The Galton Board

  http://www.randomservices.org/random/apps/GaltonBoardExperiment.html

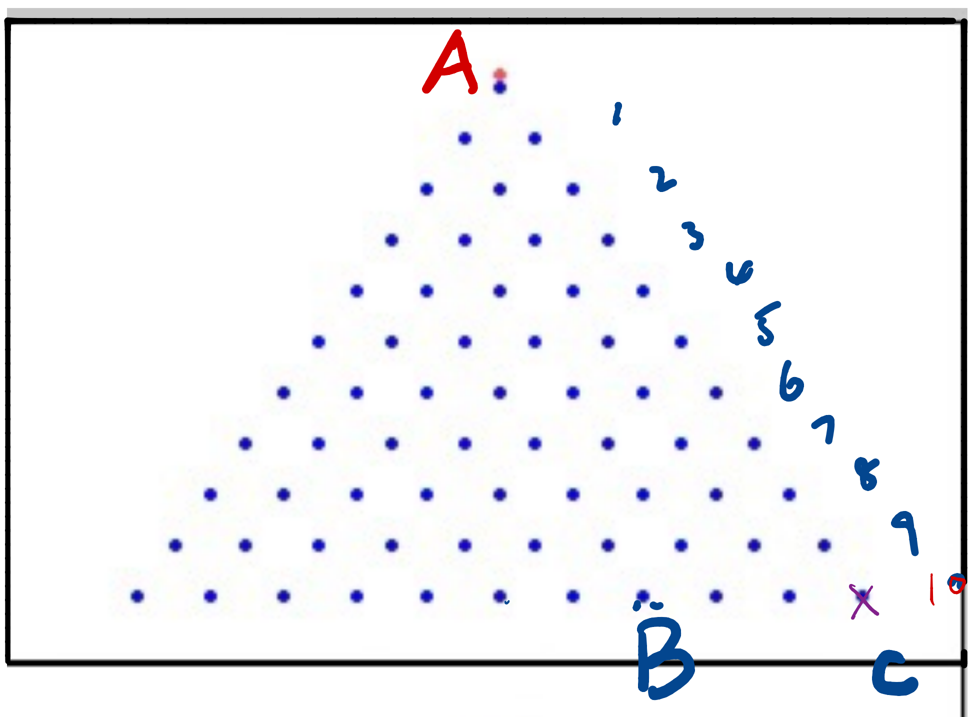
- Remember the airline problem?

How many paths that lead A → B ?

L _ _ L _ _ _ L _ _

R ? R ? R ?

10



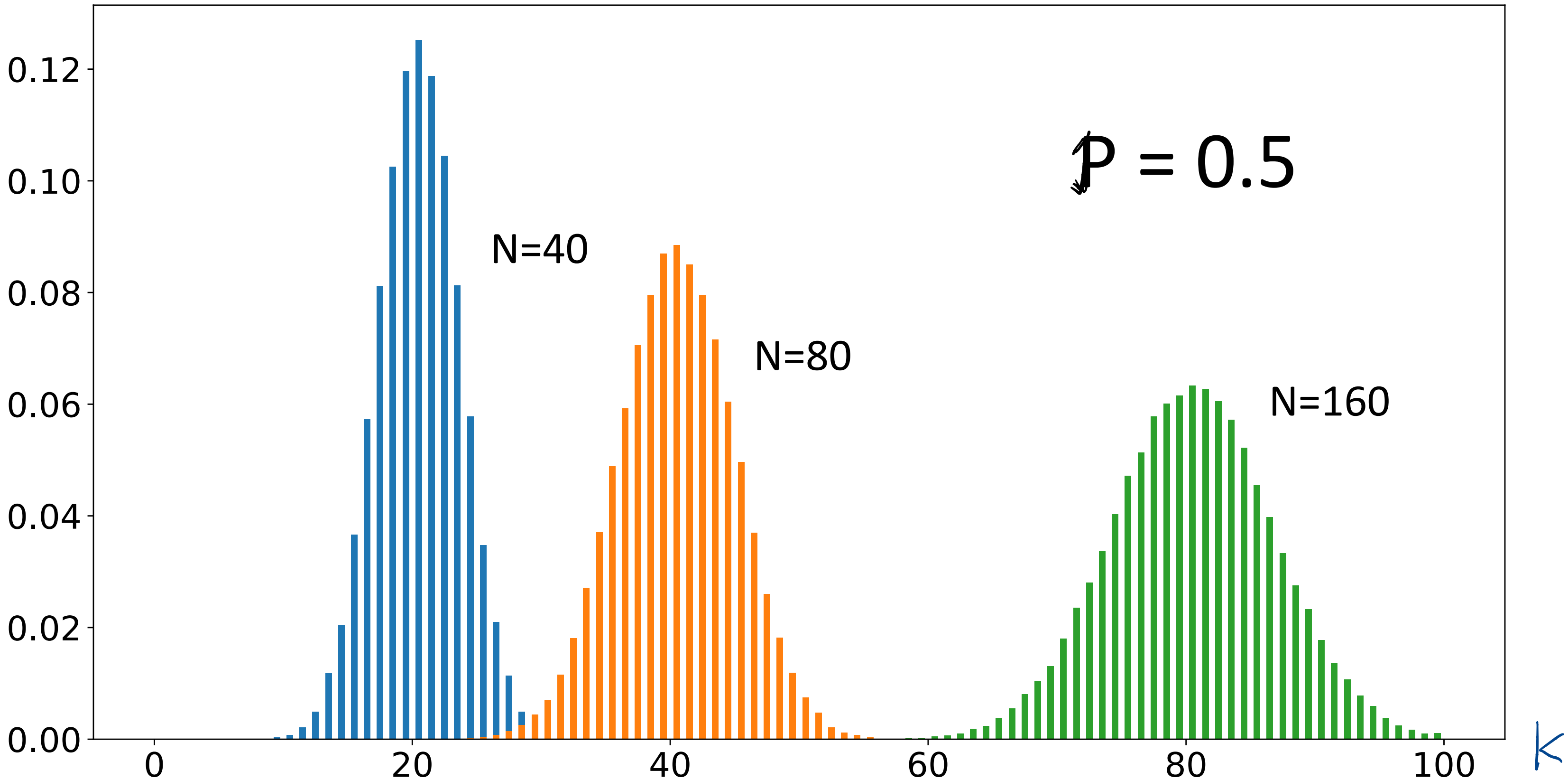
C : all "R"

B : 3 "L", 7 "R"

$$\binom{10}{3} = \binom{10}{7}$$

Binomial distribution

$P(X=k)$



Binomial distribution

- A discrete random variable X is binomial if

$$P(X=k) = \binom{N}{k} p^k (1-p)^{N-k} \quad for\ integer\ 0 \le k \le N$$

with $E[X] = Np \quad \& \quad var[X] = Np(1-p)$

$X_b = \sum_i^N X_{bei}$ $\quad X_{bei}$ iid $\quad \sum k \binom{N}{k} p^k (1-p)^{N-k}$

- Examples
  - If we roll a six-sided die N times, how many sixes we will see
  - If I attempt **N** free throws, how many points will I score
  - **What is the sum of N independent and identically distributed Bernoulli trials?**

H O H O H O O $N=7$

$P(X=3)=?$

$p=0.7$

$$\binom{7}{3} p^3 (1-p)^4$$

$$\binom{7}{3} 0.7^3 \, 0.3^4$$

Expectations of Binomial distribution

- A discrete random variable X is binomial if

$$P(X=k)=\binom{N}{k}p^k(1-p)^{N-k} \quad for\ integer\ 0 \le k \le N$$

with $E[X]=Np \quad \& \quad var[X]=Np(1-p)$

Method ①

$$\sum_x x \cdot p(x) = \sum k \binom{N}{k} p^k (1-p)^{N-k}$$

hard to solve

Method ②

$$\# \text{ heads } X = \sum_{i=1}^{N} X_{bei} \qquad X_{bei} = \begin{cases} 1 & H \\ 0 & T \end{cases}$$

$$E[X] = E\left[\sum X_{bei}\right] = \sum^{N} E[X_{bei}] = N \cdot E[X_{bei}]$$

$$= N \cdot p$$

$$(a+b)^n = \sum \binom{n}{k} a^k b^{n-k}$$

if

$a = p$

$b = 1-p$

$$1^n = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

$$1 = \sum_{k=0}^{n} P(X=k)$$

$$
\begin{aligned}
E[X] &= \sum_{k=0}^{N} k \binom{N}{k} p^{k} (1-p)^{N-k} \\
&= \sum_{k=0}^{N} \frac{k \cdot N!}{k!\,(N-k)!} p^{k} (1-p)^{N-k} \\
&= \sum_{k=1}^{N} \frac{N\,(N-1)!}{(k-1)!\,(N-k)!}\, p\, p^{k-1} (1-p)^{N-k} \\
&= N \cdot p \cdot \sum_{k=1}^{N} \frac{(N-1)!}{(k-1)!\,(N-k)!} p^{k-1} (1-p)^{N-k} \\
&= N \cdot p \cdot \sum_{k'=0}^{N-1} \frac{(N-1)!}{k'!\,(N-1-k')!} p^{k'} (1-p)^{N-1-k'} \\
&= N \cdot p \,(p + (1-p))^{N-1} \\
&= N \cdot p
\end{aligned}
$$

$$\mathrm{var}[X] = \mathrm{var}\left[\sum_{i=1}^{N} X_{bei}\right]$$

recall

$$\mathrm{var}[X+Y] = \mathrm{var}[X] + \mathrm{var}[Y]$$

if X, Y indpt

$\because X_{bei}$ are mutually indpt.

$$= \sum_{i=1}^{N} \mathrm{var}[X_{bei}]$$

$$= N \cdot p(1-p)$$

$$\mathrm{var}[X] = \sum (x - E[X])^2 p(x)$$

$$= E[X^2] - (E[X])^2$$

Binomial distribution: die example

- Let X be the number of sixes in 36 rolls of a fair six-sided die. What is P(X=k) for k =5, 6, 7

$N=36$

$p=\frac{1}{6}$

$P(X=5)=?$ $\binom{36}{5}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^{31}$

$P(X>5)=\sum_{i=6,7,\ldots 36} P(X=i)$

$1-P(X\leq 5)$

$=1-\sum_{k=0}^{5} P(X=k)$

- Calculate E[X] and var[X]

Geometric distribution

- A discrete random variable X is geometric if

$$P(X = k) = (1-p)^{k-1}p \qquad k \geq 1$$

H, TH, TTH, TTTH, TTTTH, TTTTTH,...

$k = 5$ $k = 6$

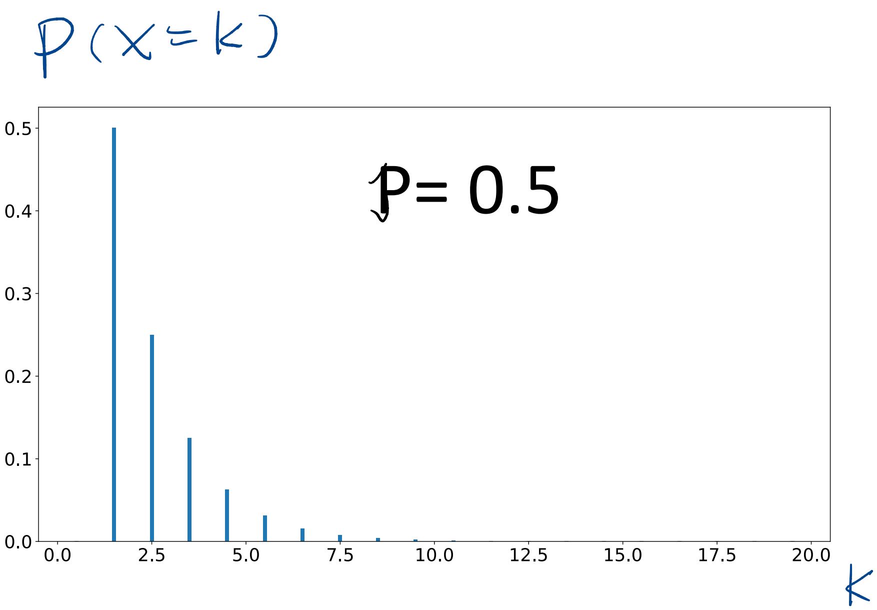
$$\overbrace{(1-p)(1-p)\cdots(1-p)}^{k-1}p$$

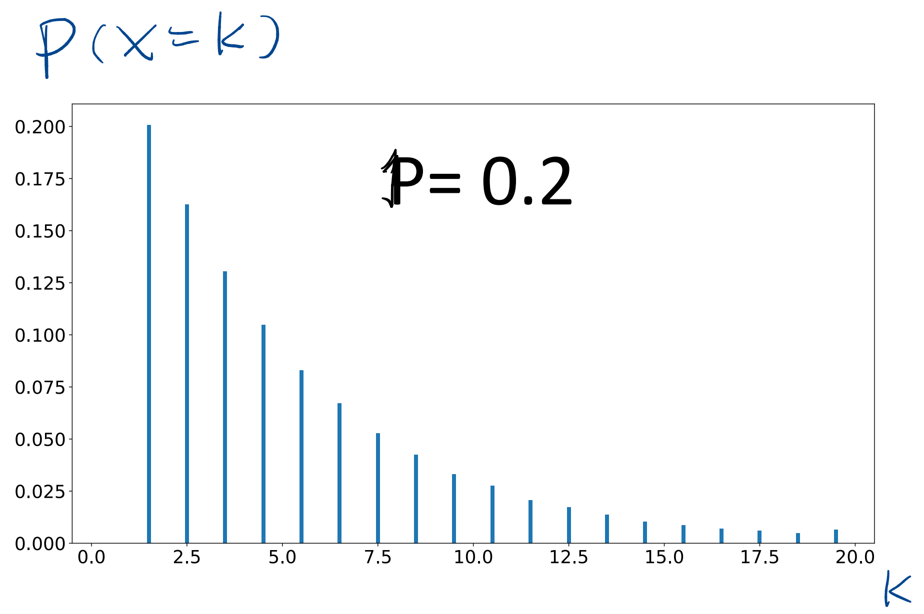
- Expected value and variance

$$E[X] = \frac{1}{p} \quad \& \quad var[X] = \frac{1-p}{p^2}$$

Geometric distribution

$$P(X = k) = (1-p)^{k-1}p \qquad k \geq 1$$





Geometric distribution

- Examples:
  - How many rolls of a six-sided die will it take to see the first 6?
  - How many Bernoulli trials must be done before the first 1?
  - How many experiments needed to have the first success?
  - Plays an important role in the **theory of queues**

Review at home

Derivation of geometric expected value

$$E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1} p$$

$$= p \sum_{k=1}^{\infty} k(1-p)^{k-1}$$

$$= \frac{p}{1-p} \sum_{k=1}^{\infty} k(1-p)^{k}$$

Review at home

Derivation of geometric expected value

$$E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

$$= p\sum_{k=1}^{\infty} k(1-p)^{k-1}$$

$$= \frac{p}{1-p}\sum_{k=1}^{\infty} k(1-p)^{k}$$

- For we have

this power series:

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}; \quad |x| < 1$$

Review at home

Derivation of geometric expected value

$$E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

$$= p\sum_{k=1}^{\infty} k(1-p)^{k-1}$$

$$= \frac{p}{1-p}\sum_{k=1}^{\infty} k(1-p)^{k}$$

$$x = 1-p$$

- For we have this power series:

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}; \quad |x| < 1$$

Review at home

Derivation of geometric expected value

$$E[X] = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

$$= p\sum_{k=1}^{\infty} k(1-p)^{k-1}$$

$$= \frac{p}{1-p}\sum_{k=1}^{\infty} k(1-p)^{k} \boxed{= \frac{1}{p}}$$

- For we have

this power series: $$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}; \quad |x| < 1$$

Review at home

Derivation of the power series

$$S(x) = \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}; \quad |x| < 1$$

Proof:

$$\frac{S(x)}{x} = \sum_{n=1}^{\infty} n x^{n-1}\, ; \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}; \quad |x| < 1$$

$$\int_0^x \frac{S(t)}{t} dt = \sum_{n=1}^{\infty} x^n = x \cdot \frac{1}{1-x} = \frac{x}{1-x}$$

$$\frac{S(x)}{x} = \left(\frac{x}{1-x}\right)'$$

$$S(x) = \frac{x}{(1-x)^2}$$

Geometric distribution: die example

- Let X be the number of rolls of a fair six-sided die needed to see the first 6. What is $P(X = k)$ for k = 1, 2?

$P(X=1) =$

$P(X=2) =$

$P(X \geq 3)$

- Calculate E[X] and var[X]

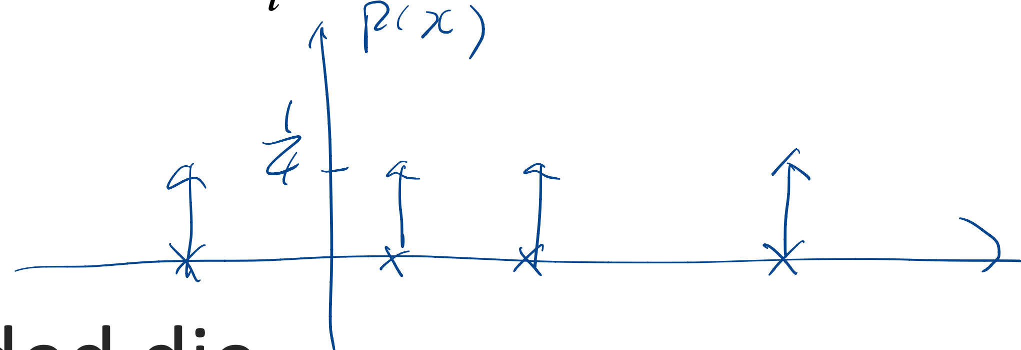
$$E[X] = \frac{1}{p} \quad \& \quad var[X] = \frac{1-p}{p^2}$$

Discrete uniform distribution

- A discrete random variable X is uniform if it takes k different values and

$$P(X = x_i) = \frac{1}{k}$$ For all x_i that X can take

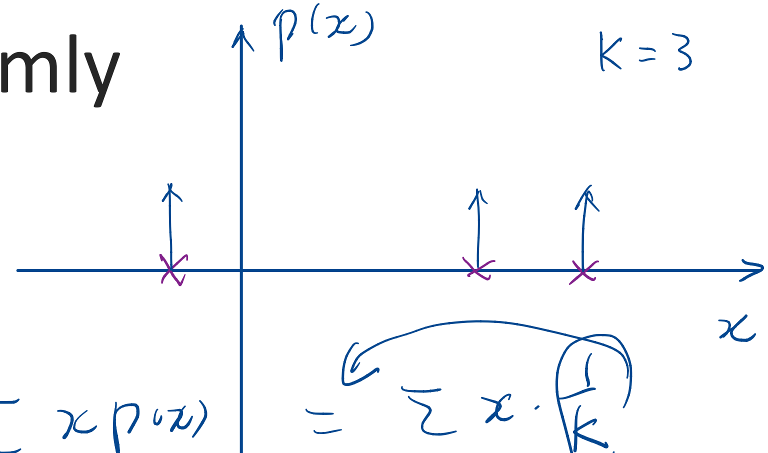
- For example:
  - Rolling a fair k-sided die
  - Tossing a fair coin (k=2)



Discrete uniform distribution

- Expectation of a discrete random variable X that takes k different values uniformly

$$E[X] = \frac{1}{k}\sum_{i=1}^{k} x_i$$



$$\sum_x x\,p(x) = \sum_x x \cdot \frac{1}{K}$$

- Variance of a uniformly distributed random variable X.

$$var[X] = \frac{1}{k}\sum_{i=1}^{k}(x_i - E[X])^2$$

Read at home

Multinomial distribution

- A discrete random variable X is Multinomial if

$$P(X_1 = n_1, X_2 = n_2, ..., X_k = n_k) = \frac{N!}{n_1!n_2!...n_k!} p_1^{n_1} p_2^{n_2} ... p_k^{n_k}$$

$$where \;\; N = n_1 + n_2 + ... + n_k$$

- The event of throwing *N* times the k-sided die to see the probability of getting n_1 X_1, n_2 X_2, n_3 X_3...n_k X_k

Read at home

Multinomial distribution

- A discrete random variable X is Multinomial if

$$P(X_1 = n_1, X_2 = n_2, \ldots, X_k = n_k) = \frac{N!}{n_1!n_2!\ldots n_k!} p_1^{n_1} p_2^{n_2} \ldots p_k^{n_k}$$

$$where \;\; N = n_1 + n_2 + \ldots + n_k$$

- The event of throwing k-sided die to see the probability of getting n_1 X_1, n_2 X_2, n_3 X_3...

ILLINOIS? $\quad \dfrac{8!}{3!2!1!1!1!}$

I L

Work at home

Multinomial distribution

- Examples
  - If we roll a six-sided die N times, how many of each value will we see?
  - This is very widely used in genetics

Work on it at home

Multinomial distribution: die example

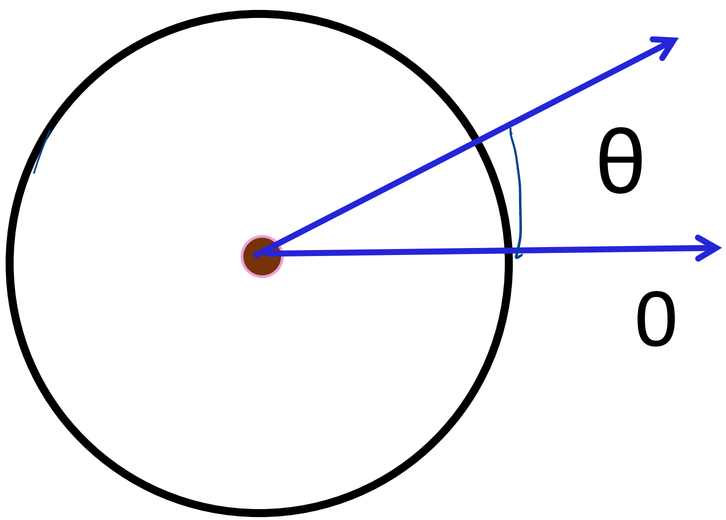
- What is the probability of seeing 1 one, 2 twos, 3 threes, 4 fours, 5 fives and 0 sixes in 15 rolls of a fair six-sided die?

Example of a continuous random variable

- The spinner

$p(a < \theta < b)$

$p(0 < \theta < 30^\circ) = \frac{30}{360} = \frac{1}{12}$



$$\theta \in (0, 2\pi]$$

$\cancel{p(\theta = \theta_0)}$

- The sample space for all outcomes is not countable

Probability density function (pdf)

- For a continuous random variable X, the probability that X=x is essentially zero for all (or most) x, so we can't define $P(X = x)$
- Instead, we define the **probability density function** (pdf) over an infinitesimally small interval dx, $p(x)dx = P(X \in [x, x + dx])$

$p(x_0) = \lim_{dx \to 0} \frac{P(x_0 < X < x_0 + dx)}{dx}$

- For $a < b$

$$\int_a^b p(x)dx = P(X \in [a, b])$$

$x \in (a, b)$ is union of $x \in (a, a+dx)$
$x \in (a+dx, a+2dx)$
...

Probability density function (pdf)

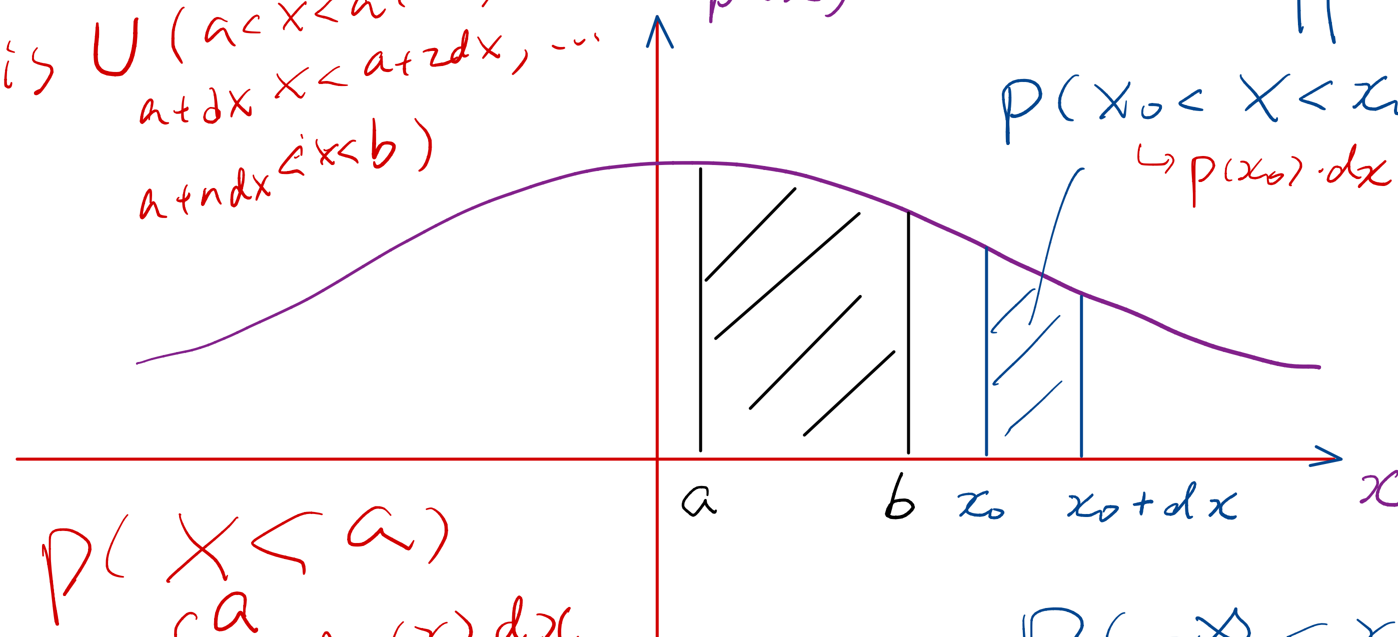
- For $a < b$

$$\int_a^b p(x)dx = P(X \in [a,b])$$

$a < X < b$

is $\cup$ ($a < X < a + dx$,
$a + dx < X < a + 2dx$, ...
$a + ndx < x < b$)

$p(x_0)dx$
$\|$
$P(X_0 < X < x_0 + dx)$
$\hookrightarrow p(x_0) \cdot dx$



dx is infinitesimal

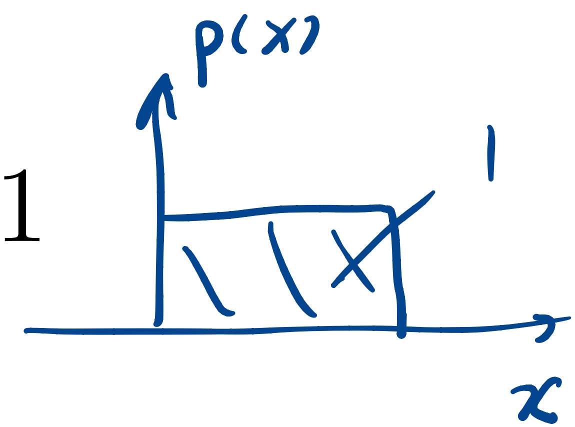
$P(X < a) = \int_{-\infty}^{a} p(x)dx$

$\underline{P(-\infty < X < \infty) = 1}$

Properties of the probability density function

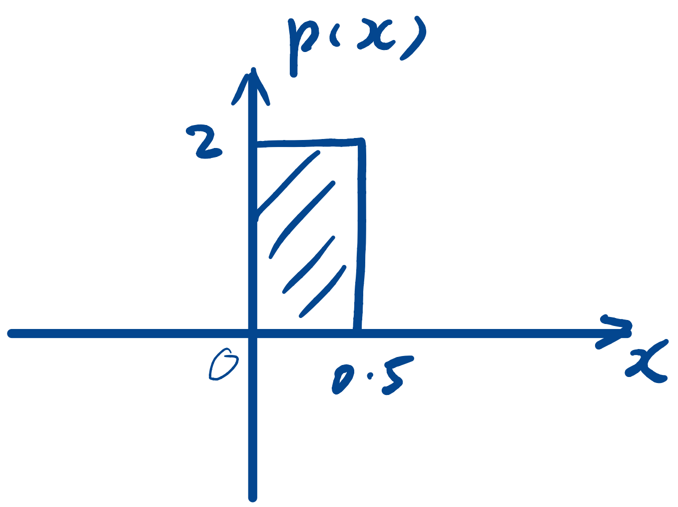
- $p(x)$ **resembles** the probability function of discrete random variables in that
  - $p(x) \geq 0 \quad$ for all x $\qquad \int_{-\infty}^{\infty} p(x)\,dx = 1$
  - The probability of X taking all possible values is 1.

$$\int_{-\infty}^{\infty} p(x)dx = 1$$



Properties of the probability density function

- $p(x)$ **differs** from the probability distribution function for a discrete random variable in that
  - $p(x)$ is not the probability that $X = x$
  - $p(x)$ can exceed 1



Assignments

- Work on Week5 material
- Next time: more classic known probability distributions

Additional References

- Charles M. Grinstead and J. Laurie Snell "Introduction to Probability"
- Morris H. Degroot and Mark J. Schervish "Probability and Statistics"

See you next time

See You!